

Theoretical Extensions and Methodological Output of the Quantum-Classical Hybrid: The Regime-Factor Purity Spectrum, the Binary Structure of the Confirmed-Breakpoint Network, and Cross-Disciplinary Predictions

——Paradigm Deepening, Testable Predictions, and Complete Deployment of the Operating System Based on the Triple Demotion

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Abstract

Tang (2026x) completed the triple demotion of the nuclear shell—the shell model was proven to be the projection of the regime-switch network in the spherical-symmetry limit by the 3.68–9.68-fold amplification of the Chow F of the five traditional magic numbers under continuous sphericity weighting (independently reproducible), the liquid drop model was proven to be the projection of the regime-switch network in the classical limit, and the independent-particle-motion axiom was proven to be incomplete. On this theoretical foundation, this paper accomplishes three independent theoretical extensions. First—the cross-disciplinary corollary of the regime-factor purity spectrum. The confirmed-breakpoint spacing $CV = 0.67$ falls between the purely discrete (particle-physics color-charge three-state $CV = 0$; chemical valence-electron equal-spacing $CV = 0$) and the purely continuous synthetic baseline (CV median = 0.689). Based on the regime-factor purities of six disciplines, the purity spectrum is formally proposed—from the fully continuous regime factors of finance to the purely discrete regime factors of particle physics and chemistry—providing a unified explanatory framework for the differentiation of demotion types: purely discrete regime factors permit constructive derivational demotion, semi-continuous regime factors permit limit-degradation derivational demotion, and fully continuous regime factors permit paradigm demotion. This purity spectrum supplies a testable predictive framework for the deployment of the Factor Hierarchy Law in a seventh discipline. The binary structure of the confirmed-breakpoint network—five confirmed breakpoints forming the full-N-axis skeleton (spacing $CV = 0.67$) and fifteen candidate breakpoints filling the inter-skeleton gaps (mean spacing 9.7)—is identified as a distinctive organizational feature of semi-continuous regime factors, standing in sharp contrast to the equal-spacing regime grids of purely discrete regime factors. Second—methodological insights into the paradigm shift. The stark contrast between the WLS continuous-weight approach and the doubly-magic-enrichment binary-screening approach (amplification of 3.68–9.68-fold versus 0.01–0.05-fold) reveals the deep physical implication that "the spherical-symmetry limit is a weight, not a region"—sphericity is continuous, and the applicability of the shell model is continuous. The cross-disciplinary methodological value of activation-operator stratification is systematically articulated for the first time: inv-family operators are sensitive to the quantum region, and raw-family operators are sensitive to the classical region—this stratification provides a direct operational guide for selecting optimal

statistical detectors in other disciplines. The uniqueness of the double-track intersection point $N \approx 70.8$ is demonstrated by an elimination-method proof using the seven-metric intersection matrix—the intersection point between the shell-correction energy ΔR^2 and the intercept/slope ratio is the only metric pair among twenty-one pairs with a clear and stable intersection characteristic; this elimination method can be generalized to other disciplines to locate quantum-classical equilibrium points. Third—the concretization of cross-disciplinary predictions and the complete deployment of the operating system. The mid-shell silence zone is proposed as a testable cross-disciplinary hypothesis for quantum-classical hybrid systems—the silence zone within nuclear physics has been confirmed by multi-operator cross-validation, and numerical simulations of a dual-signal mixing model confirm the statistical principle that "mixed signals systematically suppress the Chow F," but its applicability in condensed matter physics and quantum chemistry depends on confirmation by independent experiments. The QCTI index provides a cross-disciplinarily universal quantitative framework for the degree of quantum-classical mixing—the simplified QCTI (requiring only macroscopic-model residuals plus discrete markers, $\rho = -0.93$) can be directly deployed in any system with an analogous structure, and its applicability boundary is precisely defined as "requiring a macroscopic baseline with $R^2 > 0.9$." The complete deployment of the Factor Hierarchy Law operating system in nuclear physics—all six functional dimensions fully covered by over twenty distinct tools—and the unique methodological contributions of the nuclear-physics deployment provide a complete operational guide for the deployment of the Factor Hierarchy Law in a seventh discipline.

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1. Introduction

1.1 The Theoretical Task after the Triple Demotion

The conceptual tension between the nuclear shell model and the liquid drop model persisted for nearly nine decades. Tang (2026v) deployed the Factor Hierarchy Law to nuclear physics for the first time, performing a zero-preset regime audit on the full AME2020 nuclide dataset, locking N and Z as regime factors, discovering the $N=14$ statistical rupture, and refining six quantitative theorems. Tang (2026w) revealed the structural regularities of the blind-detected breakpoint network—the Transition-Zone Width Theorem, the Cascade Skeleton Theorem, and the Switching-Type Spectrum Trend Theorem—completing the completeness audit of the shell model.

Tang (2026x) completed the unified argument for the triple demotion. The 2D Ising exact-solution calibration established the intercept/slope ratio as a rigorous physical metric of the distance to the phase-

transition critical point (Theorem A, Spearman $\rho = 1.0000$), and dual physical verification simultaneously anchored the strict relationships of the intercept/slope ratio with fluctuation intensity and mean-field description error (two $\rho = -1.0000$). Under continuous sphericity weighting, the Chow F of the five traditional magic numbers was amplified by a factor of 3.68–9.68 (Theorem C, $\sigma = 3.0$; amplification > 1.5 for all $\sigma = 2.0$ – 5.0 , confirmed by independent-environment reproduction). The shell model was proven to be the projection of the regime-switch network in the spherical-symmetry limit, the liquid drop model was proven to be the projection of the regime-switch network in the classical limit, and the independent-particle-motion axiom was proven to be incomplete. All eight conditions for the completion of derivational demotion were satisfied.

The core argument of the triple demotion has already been completed in Tang (2026x). This paper does not repeat those arguments but rather accomplishes three independent theoretical extensions on that foundation: the cross-disciplinary corollary of the regime-factor purity spectrum, methodological insights into the paradigm shift, and the concretization of cross-disciplinary predictions together with the summary of the operating-system deployment.

1.2 Deepening and Extension Relative to the Preceding Papers

The independent contributions of this paper comprise the following seven items.

(1) **Multi-dimensional quantification of the semi-continuity of regime factors** (§2.1): The statistical characteristics of N and Z as "semi-continuous regime factors" are systematically quantified along four dimensions—the confirmed-breakpoint spacing $CV = 0.67$, the Chow F zonal CV, the local variance of the intercept/slope ratio, and cross-disciplinary benchmark comparison.

(2) **The binary structure of the confirmed-breakpoint network** (§2.1): The "skeleton-flesh" binary structure—five confirmed breakpoints forming the full-N-axis skeleton and fifteen candidate breakpoints filling the inter-skeleton gaps—is identified for the first time and positioned as a distinctive organizational feature of semi-continuous regime factors.

(3) **The cross-disciplinary corollary of the regime-factor purity spectrum** (§2.2): Based on the regime-factor purities of six disciplines, a complete purity spectrum from fully continuous to purely discrete is formally proposed, and a testable predictive framework for the differentiation of demotion types with purity is derived.

(4) **The deep physical implication of the WLS versus enrichment-zone contrast** (§3.1): The methodological insight that "the spherical-symmetry limit is a weight, not a region" is advanced to its physical root—sphericity is continuous, and the applicability of the shell model is continuous.

(5) **The cross-disciplinary methodological value of activation-operator stratification** (§3.2): The selective sensitivity of inv-family operators to the quantum region and raw-family operators to the classical region is systematically articulated for the first time, providing an operational guide for selecting optimal statistical detectors in other disciplines.

(6) **The elimination-method proof of the uniqueness of the double-track intersection point (§4.1):** The seven-metric intersection-matrix analysis is positioned as an elimination-method proof—the intersection point between the shell-correction energy ΔR^2 and the intercept/slope ratio is the only metric pair among twenty-one pairs with a clear and stable intersection characteristic.

(7) **The concretization of cross-disciplinary predictions and the summary of the operating-system deployment (§§4–5):** Specific cross-disciplinary deployment protocols and falsification conditions are provided for the mid-shell silence zone, the double-track intersection point, and the QCTI, and the complete deployment of the Factor Hierarchy Law operating system in nuclear physics is systematically summarized.

It should be made clear that the core empirical support for the paradigm shift comes from Theorem C (WLS all-five-magic-number amplification, independently reproducible) and Theorem B (seven-metric evidence matrix, $\rho = -0.95$) of Tang (2026x). This paper does not repeat those arguments but rather provides new independent contributions on that foundation.

1.3 Core Conclusions of the Preceding Papers

To facilitate independent reading, **Table 0** summarizes the core theorems and theoretical contributions of the four papers.

Table 0: Core Theorems and Theoretical Contributions of the Four Papers

Paper	Core Contribution	Core Theorems	Argument Type
Tang (2026v)	Empirical discoverer	Theorems 1–6 (six theorems)	Statistical discovery
Tang (2026w)	Network revealer	Theorems 7–9 (three theorems)	Structural revelation
Tang (2026x)	Triple-demotion completer	Theorems A–F (six theorems)	Theoretical unification
This paper	Theoretical extension and methodological output	Theorem F (complete statement)	Theoretical extension and summary

2. The Regime-Factor Purity Spectrum and the Binary Structure of the Confirmed-Breakpoint Network

Tang (2026x) has already systematically compared the independent-particle-motion paradigm with the quantum-classical hybrid paradigm along nine dimensions and completed the core argument of the triple demotion (Theorems A–F). This chapter does not repeat those arguments but rather provides independent empirical support and theoretical extension for the paradigm shift from three dimensions: the multi-dimensional

quantification of the semi-continuity of regime factors, the binary structure of the confirmed-breakpoint network, and the cross-disciplinary corollary of the purity spectrum.

2.1 Multi-Dimensional Quantification of the Semi-Continuity of Regime Factors and the Binary Structure of the Confirmed-Breakpoint Network

The "purity" of a regime factor is the key to distinguishing among demotion types. Purely discrete regime factors (color charge in particle physics, valence-electron number in chemistry) permit derivational demotion; fully continuous regime factors (the China-U.S. interest-rate spread in finance) permit only paradigm demotion. N and Z occupy a "semi-continuous" position—they exhibit strong regime-factor function near magic numbers, but large regions between magic numbers lack confirmed breakpoints.

Table 1 quantifies this semi-continuity along four dimensions. The confirmed-breakpoint spacing $CV = 0.67$ falls between the purely discrete ($CV = 0$) and the purely continuous synthetic baseline (CV median = 0.689, 95% range [0.247, 1.275]). It should be candidly acknowledged that $CV = 0.67$ is calculated from only four spacing data points (66, 19, 21, 16) and is sensitive to extremely small samples. The nuclear-physics $CV = 0.67$ is statistically significantly larger than zero (it cannot be purely discrete), but the difference from the purely continuous median of 0.689 does not reach statistical significance—the nuclear-physics CV falls within the 95% range of the purely continuous synthetic baseline, meaning that the CV alone cannot completely distinguish the "semi-continuous" of nuclear physics from the "purely continuous" of a purely random distribution. The positioning as "semi-continuous" requires the joint support of the binary structure of the confirmed-breakpoint network, the Chow F zonal CV , and other dimensions.

The Chow F zonal CV is less than 1 in all three nuclear regions, indicating that a stable zonal regime exists within each nuclear region rather than random fluctuations. The local variance of the intercept/slope ratio reaches 684.9 in the heavy-nucleus region, far exceeding the 109.6 in the light-nucleus region and the 111.5 in the mid-shell region, indicating that the "purity" of the regime factor does not vary monotonically along the N-axis—the variation in regime-switch types is greatest in the heavy-nucleus region, which may point to the action of other hidden regime factors in that region (such as deformation parameters or pairing strength), or to the weakening of the regulatory function of the regime factors N and Z in the heavy-nucleus region due to the dominance of classical liquid-drop behavior.

Table 1: Multi-Dimensional Quantification of Regime-Factor Purity

Metric	Light-Nucleus Region ($N \leq 50$)	Mid-Shell Region ($50 < N \leq 100$)	Heavy-Nucleus Region ($N > 100$)	Overall
Confirmed-breakpoint spacing CV	—	—	—	0.67
Chow F zonal CV	0.32	0.17	0.47	—
Intercept/slope ratio	109.6	111.5	684.9	—

local variance				
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As cross-disciplinary benchmark comparisons, the particle-physics color-charge three-state spacing $CV = 0$ (purely discrete, strictly equal spacing), and the chemical valence-electron equal-spacing regime grid $CV = 0$ (purely discrete, spacing = 1, $CV = 0$). The nuclear-physics $CV = 0.67$ falls between the purely discrete and the purely continuous; the positioning as "semi-continuous" holds in the cross-disciplinary frame of reference.

The binary structure of the confirmed-breakpoint network—five confirmed breakpoints (spacings 66, 19, 21, 16; $CV = 0.67$) and fifteen candidate breakpoints (mean spacing 9.7)—is a distinctive organizational feature of semi-continuous regime factors. The confirmed breakpoints form the full-N-axis skeleton ($23 \rightarrow 89 \rightarrow 108 \rightarrow 129 \rightarrow 145$), covering the complete range from the light-nucleus region to the superheavy-nucleus region; the candidate breakpoints fill the gaps between the skeleton elements, constituting finer secondary structures. This "skeleton-flesh" binary network stands in sharp contrast to the equal-spacing regime grids of the particle-physics color-charge three states (strictly equal spacing, $CV = 0$) and the chemical valence-electron eight states (strictly equal spacing, $CV = 0$)—purely discrete regime factors lack this binary structure, and fully continuous regime factors also lack it. The binary structure is a statistical organizational feature of semi-continuous regime factors.

2.2 The Cross-Disciplinary Corollary of the Regime-Factor Purity Spectrum

Based on the above analysis, this paper formally proposes the **Regime-Factor Purity Spectrum**—from the fully continuous regime factors of finance (China-U.S. interest-rate spread, $CV \gg 1$) through the semi-continuous regime factors of nuclear physics (N and Z, $CV = 0.67$) to the purely discrete regime factors of particle physics and chemistry (color-charge three-state $CV = 0$; valence-electron equal-spacing $CV = 0$)—constituting a complete spectrum from the continuous to the discrete.

The core corollary of the purity spectrum is that the purity of the regime factor determines the degree of completion that the demotion argument can attain in that discipline. Purely discrete regime factors permit constructive derivational demotion—the mathematical structure of the traditional theory is rigorously derived from the regime-state structure via group theory (particle physics: three color-charge states $\rightarrow SU(3)$; chemistry: eight valence-electron states $\rightarrow s, p, d, f$). Semi-continuous regime factors permit limit-degradation derivational demotion—the traditional theory is proven to be incomplete, and its core predictions are recovered in specific limits (nuclear physics: spherical-symmetry limit $\rightarrow$ shell-model recovery; classical limit $\rightarrow$ liquid-drop-model recovery). Fully continuous regime factors permit paradigm demotion—the traditional paradigm is proven, in the context of methodology, to be a manifestation of regime switching under specific institutional conditions (finance: China-U.S. interest-rate-spread regime switch $\rightarrow$ interest-rate-parity paradigm demotion).

This purity spectrum supplies a testable predictive framework for the deployment of the Factor Hierarchy Law in a seventh discipline: if the regime factor of the new discipline is purely discrete ($CV \approx 0$), it is predicted

that constructive derivational demotion can be achieved in that discipline; if it is semi-continuous ($CV \approx 0.5-0.7$), it is predicted that limit-degradation derivational demotion can be achieved; if it is fully continuous ($CV \gg 1$), it is predicted that paradigm demotion can be achieved. The testability of this predictive framework lies in the fact that, after deploying the Factor Hierarchy Law in a new discipline, the purity of the regime factor can be independently calculated from the confirmed-breakpoint spacing CV, and the degree of completion of the demotion argument can be assessed by the eight-condition evaluation—the correspondence between the two constitutes a falsifiable cross-disciplinary prediction.

It should be candidly acknowledged that the current purity spectrum is based on empirical induction from only six disciplines—three data points (constructive type for particle physics and chemistry, limit-degradation type for nuclear physics, paradigm type for finance) forming a spectrum constitute a directional observation rather than an established regularity. The six disciplines exhibit only three demotion types, and the three types correspond to three purities—this correspondence has not yet been observed with counterexamples at the same purity level. If a discipline with a purely discrete regime factor is discovered in the future that can only achieve paradigm demotion, the predictive framework of the purity spectrum will be falsified. The deployment in a seventh discipline is the genuine test of this predictive framework.

3. Methodological Insights into the Paradigm Shift

Tang (2026x) completed the core argument for the paradigm shift. This chapter does not repeat those arguments but rather provides independent methodological insights into this paradigm from two dimensions: the deep physical implication of the WLS versus enrichment-zone contrast and the cross-disciplinary methodological value of activation-operator stratification.

3.1 The WLS versus Enrichment-Zone Contrast—"The Spherical-Symmetry Limit Is a Weight, Not a Region"

Table 2 contrasts the results of the continuous-sphericity-weight WLS approach with those of the doubly-magic-enrichment-zone analysis. The WLS approach amplifies the Chow F of the five magic numbers by a factor of 3.68–9.68 (Tang, 2026x, Theorem C, independently reproducible, robust for $\sigma = 2.0-5.0$), whereas the doubly-magic enrichment zone (taking a subset within ± 3 of a magic number) yields amplification factors of only 0.01–0.05—far from being amplified, the Chow signal is instead drastically attenuated.

Table 2: Comparison of WLS Amplification and Doubly-Magic-Enrichment-Zone Amplification

Magic Number	WLS Amplification Factor ($\sigma = 3.0$)	Enrichment-Zone Amplification Factor	Methodological Difference
N=20	3.68	0.02	WLS continuous weight

N=28	6.91	0.01	WLS continuous weight
N=50	6.86	0.02	WLS continuous weight
N=82	6.84	0.05	WLS continuous weight
N=126	9.68	0.04	WLS continuous weight

This contrast reveals the deep physical implication that "the spherical-symmetry limit is a weight, not a region": sphericity is continuous, and the applicability of the shell model is continuous. "Spherical nuclides" are not a special group of nuclides that can be isolated by binary screening—sphericity takes its maximum value near magic numbers and smoothly decays with increasing distance from a magic number. The applicability of the shell model also smoothly decays with it—in the spherical-symmetry limit (sphericity weight $\rightarrow 1$), the shell model is exact; as the sphericity weight decreases, the degree of approximation of the shell model continuously declines. This stands in sharp contrast to the traditional view of the shell model ("nuclides near magic numbers are spherical")—the traditional view is binary; the discovery of the Factor Hierarchy Law is continuous.

It should be candidly acknowledged that the sample size of the enrichment zone is far smaller than that of the full sample, and the attenuation of the Chow F may partly arise from the reduction in test power due to the reduced sample size. This alternative explanation does not diminish the contrastive value of the WLS versus enrichment-zone comparison—the methodological insight that "the spherical-symmetry limit must be approached under continuous weighting" does not depend on the specific mechanism of the attenuation in the enrichment zone.

3.2 The Cross-Disciplinary Methodological Value of Activation-Operator Stratification

Tang (2026v) discovered that activation operators automatically stratify into two groups—all three inv-family operators lock onto $N=14$, and all four raw-family operators lock onto $N=30-32$. Tang (2026x) interpreted this stratification as a physical manifestation of differences in the quantum-classical mixing ratio. This paper further excavates the cross-disciplinary methodological value of this discovery.

The inv transformation amplifies the variance differences in the light-nucleus region (high quantum-effect region, extremely small intercept/slope ratio), making inv-family operators highly sensitive to sharp ruptures near quantum phase-transition points; the raw transformation preserves the global variance structure, making raw-family operators produce a maximal response to trend ruptures in the classically smooth region. This selective sensitivity has received mathematical verification on the 2D Ising exact solution— χ is completely silent under the Chow test ($F = 16.8$), but $1/\chi$ activates it to a magnitude of $\sim 10^{21}$ (Tang, 2026k).

Activation-operator stratification provides a cross-disciplinary application guide for the "activation operator" tool within the operating system of the Factor Hierarchy Law: if the goal is to detect sharp ruptures near a quantum phase-transition point, operators sensitive to the quantum region (such as the inv transformation) should be used; if the goal is to detect trend ruptures in a classically smooth region, operators sensitive to the

classical region (such as the raw transformation) should be used. This guide can directly assist other disciplines in selecting optimal statistical detectors.

3.3 Shell-Correction Compensation Traces—The Logical Chain of the "Compensation Hypothesis"

The shell-correction compensation traces (Evidence 5) were reported in Tang (2026w) and cited in Tang (2026x). This section is a systematic summary of findings already reported in Tang (2026w) and Tang (2026x) and does not constitute an independent new discovery. This logical chain comprises five links: (1) three tools consistently avoid the shell-closure positions (MWU $p = 0.0487$); (2) the no-shell-correction liquid-drop-model residual test rules out the alternative explanation of "poor local data quality"; (3) the cross-regime rupture ratio of the mass covariance is 22.31 on the same side versus across the boundary; (4) all seven columns $\times$ nine boundaries of the reaction-energy errors exhibit systematic high/low ratio changes; (5) the shell-correction term peaks in the mid-shell region—"maximum mixing requires maximum compensation." These five links constitute a multi-level verification chain for the "compensation hypothesis," logically independent yet convergent in their conclusions.

4. Cross-Disciplinary Methodological Tools of the Quantum-Classical Hybrid

This chapter elevates three structures already discovered in Tang (2026x)—the double-track intersection point, the mid-shell silence zone, and the QCTI index—into methodological tools that can be deployed across disciplines, and supplements argumentative details not fully developed in Tang (2026x).

4.1 The Elimination-Method Proof of the Uniqueness of the Double-Track Intersection Point

The double-track intersection point $N \approx 70.8$ is one of the core structures discovered by Tang (2026x). The shell-correction energy ΔR^2 and the intercept/slope ratio intersect exactly at this point, with a spectrum value of 0.407, close to the theoretically predicted 0.5. To further verify the uniqueness of this intersection, Tang (2026x) paired the seven metrics of the seven-metric evidence matrix pairwise, generating twenty-one pairs of intersection analyses, and found that the intersection point between the shell-correction energy ΔR^2 and the intercept/slope ratio is the only metric pair with a clear and stable intersection characteristic.

This paper positions this analysis as an **elimination-method proof**: among the twenty-one candidate metric pairs, those that do not satisfy the criterion of a "clear and stable intersection" are eliminated one by one, and the sole remaining metric pair constitutes the mathematical basis of the double-track intersection point. This elimination method can be generalized to other disciplines—by screening among multiple candidate metric

pairs through an intersection matrix to identify the unique metric pair with a stable intersection characteristic, the quantum-classical equilibrium point can be located.

The methodological essence of the double-track intersection point is that two independent statistics—one primarily responsive to quantum effects (the shell-correction energy ΔR^2) and the other primarily responsive to classical effects (the intercept/slope ratio)—intersect exactly at the quantum-classical equilibrium point. This intersection is not artificially constructed but emerges automatically from two completely independent statistics. In any quantum-classical hybrid system, as long as two independent statistics can be defined—one measuring the intensity of quantum effects and the other measuring the intensity of classical effects—one can scan the control-parameter space, search for their intersection point, and locate the quantum-classical equilibrium point.

4.2 The Mid-Shell Silence Zone as a Testable Cross-Disciplinary Hypothesis

The mid-shell silence zone is one of the important new structures discovered by Tang (2026x). In nuclear physics, the activation-operator Chow F drops to 6.8 in the mid-shell region, only 1/15 of the light-nucleus region's 105.4, with the Bootstrap 95% CI fully separated, and the silence determination of the six operators is completely consistent (Kendall's W far beyond the human-eye limit).

This paper proposes the mid-shell silence zone as a **testable cross-disciplinary hypothesis** for quantum-classical hybrid systems. At present, this hypothesis rests on two levels of argument. The first level (verified)—the silence zone within nuclear physics has been confirmed by multi-operator cross-validation, and numerical simulations of a dual-signal mixing model confirm the statistical principle that "mixed signals systematically suppress the Chow F." The second level (to be verified)—the applicability of this principle in condensed matter physics and quantum chemistry depends on whether functionally equivalent quantum-classical competition mechanisms exist in those systems. The "universality" of the silence zone is positioned as "the conclusion reached after this principle has been independently verified in those systems," not as "a cross-disciplinary regularity already established at present."

In condensed matter physics, the level-spacing statistics of quantum-dot arrays obey random-matrix theory in the quantum limit (low temperature, small dot number) and Poisson statistics in the classical limit (high temperature, large dot number)—in the quantum-classical transition region, the competition between the two statistics may lead to a systematic suppression of the Chow signal of conductance fluctuations. The verification method is universal: deploy multi-operator Chow tests and test whether a silence zone with significantly consistent Kendall's W exists. If no silence zone is found in three or more quantum-classical hybrid systems, the universality of the silence zone is restricted to the domain of nuclear physics.

4.3 The Cross-Disciplinary Operational Definition and Applicability Boundary of the QCTI

The QCTI is a composite index constructed by Tang (2026x), integrating three independent metrics: the shell-correction energy ΔR^2 , the binary-marker ΔR^2 , and the intercept/slope ratio. The Spearman ρ of the QCTI across the full N-axis is -0.9502 . The simplified QCTI (using only the shell-correction energy ΔR^2 and the binary-marker ΔR^2) yields $\rho = -0.9261$; the two constituent metrics are each independently and significantly correlated with N ($\rho = -0.8736$ and $\rho = -0.5677$, respectively).

The construction method of the simplified QCTI possesses cross-disciplinary universality—requiring only two metrics that can be directly deployed in any system with an analogous structure: macroscopic-model residuals (corresponding to the shell-correction energy ΔR^2) and discrete markers (corresponding to the binary-marker ΔR^2). It should be made clear that the universality of the simplified QCTI depends on the existence of a high-precision macroscopic baseline. In nuclear physics, the liquid-drop model $R^2 = 0.96$ ensures the validity of the simplified QCTI—the macroscopic model already accurately describes the vast majority of the variance, and what remains in the residuals is principally the quantum shell effect. If a quantum-classical hybrid system lacks a macroscopic model with $R^2 > 0.9$, the simplified QCTI may not be deployable. This is not a defect of the theory but a precise delineation of the theory's applicability boundary—it tells subsequent researchers when this tool can be used and when it cannot.

4.4 Methodological Implications of Nuclear Physics for Cross-Disciplinary Research

The complete deployment of the Factor Hierarchy Law operating system in nuclear physics provides a methodological template for other quantum-classical hybrid systems. The three-step core logic of the MRSD algorithm—input objective data $\rightarrow$ perform structural optimization $\rightarrow$ output the regime division-of-labor matrix—has been fully implemented in nuclear physics. The discovery of activation-operator stratification reveals the selective sensitivity of different operators to quantum and classical regions, which can guide other disciplines in selecting optimal statistical detectors. The construction method of the seven-metric evidence matrix can be directly transferred to any quantum-classical hybrid system to test whether systematic zonal characteristics exist.

5. Testable Theoretical Predictions

Tang (2026x) has already proposed seven testable predictions, covering predictions internal to nuclear physics (QCTI trend in the superheavy-nucleus region, shell-correction decay, secondary breakpoints, confirmation of candidate breakpoints) and cross-disciplinary predictions (condensed-matter silence zone, quantum-chemistry double-track intersection point, r-process waiting points). Each prediction is accompanied

by a specific verification path and a falsification condition (see Tang, 2026x, Table 7 for details). This paper does not repeat the full content of those predictions but supplements the verification details of the cross-disciplinary predictions.

Table 3 summarizes the verification details and potential application domains of the cross-disciplinary predictions.

Table 3: Verification Details and Potential Application Domains of the Cross-Disciplinary Predictions

Prediction	Verification Platform / Method	Quantum Metric	Classical Metric	Potential Application Domain
Condensed-matter silence zone	Quantum-dot arrays, cold-atom optical lattices	Coulomb-blockade strength, superfluid order parameter	Classical conductance, atom-cloud density	Quantum-device design, quantum-phase-transition detection
Quantum-chemistry double-track intersection point	CASSCF/MRCI calculations	Non-adiabatic coupling strength	Nuclear kinetic energy	Photochemical reaction mechanisms, excited-state dynamics
r-process waiting points	Precision nuclear mass measurements + Bai-Perron	Shell-correction energy ΔR^2	Liquid-drop model R^2	Nuclear astrophysics, origin of the elements

For the condensed-matter silence zone, specific verification platforms include semiconductor quantum-dot arrays and cold-atom optical lattices. In semiconductor quantum-dot arrays, one can measure conductance fluctuations at different temperatures and deploy multi-operator Chow tests on the quantum-dot-number axis. The quantum limit corresponds to low temperatures and small dot numbers; the classical limit corresponds to high temperatures and large dot numbers. For the quantum-chemistry double-track intersection point, the specific computational approach involves the non-adiabatic coupling strength and the nuclear kinetic energy on the potential energy surfaces of molecular excited states. For the r-process waiting points in nuclear astrophysics, the specific observational path involves precision mass measurements of nuclei at $N = 137, 145,$ and 150 on the r-process path.

The falsification conditions for the seven predictions have been fully given in Tang (2026x, Table 7). The falsification thresholds for the cross-disciplinary predictions are as follows. Condensed-matter silence zone: if no silence zone with significantly consistent Kendall's W is found in three or more quantum-classical hybrid systems, the universality of the silence zone is restricted to the domain of nuclear physics. Quantum-chemistry double-track intersection point: if the non-adiabatic coupling strength and the nuclear kinetic energy have no

intersection point on the excited-state potential energy surface, the universality of the double-track intersection-point method is restricted. r-process waiting points in nuclear astrophysics: if Bai-Perron detects no significant breakpoints at $N = 137, 145, \text{ and } 150$ (BIC improvement < 10), the interpretation of the r-process waiting points by the regime-switch network is falsified.

6. Complete Deployment of the Factor Hierarchy Law Operating System in Nuclear Physics

6.1 Complete Coverage of the Six Functional Dimensions

The operational implementation of the Factor Hierarchy Law is the Testability Norms, comprising six functional dimensions. Across the nuclear-physics series of papers, these six dimensions have been completely covered by over twenty distinct tools.

Search completeness: exhaustive testing of one- to three-factor combinations of eighteen factors, dual hard constraints of $VIF < 5$ and $VIF < 10$, BMA weight collapse to 1.0000, Lasso + RF + MI + BMA quadruple cross-validation. **Modulation relationship:** interaction-effect test ($\eta > 10^6$), threshold interaction model (locking $N=25$ as the optimal threshold), GLS coefficient sign reversal, decay-mode interaction effects (all four decay modes $p < 10^{-23}$). **Structural breakpoints:** Chow $\rightarrow$ SupF $\rightarrow$ CUSUM $\rightarrow$ Bai-Perron $\rightarrow$ PELT $\rightarrow$ Bayesian segmentation $\rightarrow$ MOSUM $\rightarrow$ kernel change-point $\rightarrow$ HMM $\rightarrow$ seven activation operators—a total of over ten distinct tools, covering the complete detection chain of "mean $\rightarrow$ regression $\rightarrow$ local $\rightarrow$ nonlinear $\rightarrow$ hidden state." **Causal direction:** conditional independence, regression discontinuity design, instrumental variable method, DAG causal network, no-shell-correction liquid-drop-model test. **Problem completeness:** 5W2H, PRISMA (100% coverage), PICO ($N=14$ intercept jump of -245.98 keV), SWOT. **Process closure:** PDCA, multi-level defense tests (detrended permutation test, multi-tool shell-closure distribution test, random-breakpoint-set control test, internal-breakpoint deletion test, no-shell-correction model test).

6.2 Unique Methodological Contributions of the Nuclear-Physics Deployment

The nuclear-physics deployment contributes three unique components to the methodological system of the Factor Hierarchy Law. The **confirmed/candidate breakpoint dichotomy** provides an operational scheme for balancing statistical-significance screening and physical-structure preservation. The **multi-level defense test system** provides a methodological template for the deployment of the Factor Hierarchy Law in future new disciplines. The **statistical characterization toolkit for the quantum-classical hybrid**—intercept/slope ratio + QCTI + double-track intersection point + mid-shell silence zone—provides a complete statistical-characterization toolbox for any quantum-classical hybrid system.

The cross-disciplinary deployment record of the Factor Hierarchy Law operating system across six disciplines has already been systematically summarized in Tang (2026r). The unique contribution of nuclear physics, as the sixth discipline to be deployed, lies in being the first to automatically identify and quantitatively characterize this transition within the quantum-classical transition zone.

6.3 Mathematical and Physical Achievements

The nuclear-physics series of papers has left behind a systematic body of mathematical and physical achievements.

Mathematical achievements: the strictly monotonic mapping of the intercept/slope ratio to the distance from the critical point ($\rho = \pm 1.0000$); the QCTI index ($\rho = -0.95$, simplified version $\rho = -0.93$); the spectrum-value self-consistency check ($\rho = -0.82$); the Transition-Zone Width Theorem ($W_{\text{cross}} = \sum W_{\text{internal}}$); the statistical formulation of spherical-symmetry limit recovery (WLS amplification of 3.68–9.68-fold, robust for $\sigma = 2.0$ –5.0, independently reproducible).

Physical achievements: the quantum-classical hybrid as the statistical organizing principle of the nuclear shell; the unified argumentative framework of the triple demotion (shell model $\rightarrow$ spherical-symmetry limit; liquid drop model $\rightarrow$ classical limit; independent-particle-motion axiom $\rightarrow$ incompleteness); the regime-factor purity spectrum—a continuous spectrum from fully continuous to purely discrete, with purity determining the degree of completion attainable by the demotion argument; the binary structure of the confirmed-breakpoint network; the mid-shell silence zone as a testable cross-disciplinary hypothesis for quantum-classical competition; the double-track intersection point as a general method for locating the quantum-classical transition zone.

Testable predictions: seven testable predictions (four internal to nuclear physics plus three cross-disciplinary), each with a clear verification path and a falsification condition.

7. Discussion and Conclusion

7.1 The Independent Contributions of the Theoretical Extension

On the foundation of the triple demotion completed by Tang (2026x), this paper has accomplished independent theoretical extensions in three directions. First, the regime-factor purity spectrum—building on the regime-factor purities of six disciplines, a predictive framework for demotion types has been constructed. The binary structure of the confirmed-breakpoint network has been identified as a distinctive organizational feature of semi-continuous regime factors. Second, methodological insights into the paradigm shift—the WLS versus enrichment-zone contrast has revealed the deep physical implication that "the spherical-symmetry limit is a weight, not a region"; activation-operator stratification has provided a guide for selecting statistical detectors in

other disciplines; and the elimination-method proof of the double-track intersection point can be generalized to other disciplines. Third, the concretization of cross-disciplinary predictions and the summary of the operating-system deployment—specific cross-disciplinary deployment protocols and falsification conditions have been provided for the mid-shell silence zone, the double-track intersection point, and the QCTI, and the complete deployment of the Factor Hierarchy Law operating system in nuclear physics has been systematically summarized.

The core empirical support for the paradigm shift comes from Theorem C and Theorem B of Tang (2026x). The contribution of this paper lies in providing independent empirical details, methodological insights, and cross-disciplinary extensions for this paradigm from new angles—the regime-factor purity spectrum, the binary structure of the confirmed-breakpoint network, activation-operator stratification, and so forth.

7.2 The Cross-Disciplinary Significance of the Purity Spectrum

The regime-factor purity spectrum provides a unified explanatory framework for the differentiation of demotion types and supplies a testable prediction for the deployment of the Factor Hierarchy Law in a seventh discipline. It should be candidly acknowledged that the current purity spectrum is based on empirical induction from only six disciplines—the six disciplines exhibit only three demotion types, and the three types correspond to three purities; this correspondence has not yet been observed with counterexamples at the same purity level. The deployment in a seventh discipline is the genuine test of this predictive framework.

7.3 Complete Deployment of the Factor Hierarchy Law in a Sixth Discipline

From finance to nuclear physics—six disciplines, the same operating system. The unique contribution of nuclear physics lies in the fact that the Factor Hierarchy Law has, for the first time, automatically identified and quantitatively characterized this transition within the quantum-classical transition zone. The organizing principles traditionally regarded as "fundamental" are all approximate projections of regime switching under specific conditions.

7.4 The Final Positioning of the Four Papers

Table 4 summarizes the progressive logic of the four papers.

Table 4: The Progressive Logic of the Four Papers

Paper	Core Contribution	Core Theorems	Argument Type
Tang (2026v)	Empirical discoverer	Theorems 1–6 (six theorems)	Statistical discovery
Tang (2026w)	Network revealer	Theorems 7–9 (three theorems)	Structural revelation

Paper	Core Contribution	Core Theorems	Argument Type
Tang (2026x)	Triple-demotion completer	Theorems A–F (six theorems)	Theoretical unification
This paper	Theoretical extension and methodological output	Theorem F (complete statement)	Theoretical extension and summary

7.5 Conclusion

On the foundation of the triple demotion completed by Tang (2026x), this paper has accomplished three independent theoretical extensions. The regime-factor purity spectrum—a continuous spectrum from fully continuous to purely discrete—provides a unified explanatory framework for the differentiation of demotion types and supplies a testable predictive framework for the deployment of the Factor Hierarchy Law in a seventh discipline. The binary structure of the confirmed-breakpoint network has been identified as a distinctive organizational feature of semi-continuous regime factors. The WLS versus enrichment-zone contrast has revealed the deep physical implication that "the spherical-symmetry limit is a weight, not a region." Activation-operator stratification provides an operational guide for selecting optimal statistical detectors in other disciplines. The elimination-method proof of the double-track intersection point can be generalized to other disciplines to locate quantum-classical equilibrium points. The seven testable predictions provide adjudicative tests for future experiments. The complete deployment of the Factor Hierarchy Law operating system in nuclear physics provides a complete operational guide for its deployment in a seventh discipline.

The completion of the four papers of the nuclear-physics series—the empirical discoverer (Tang, 2026v), the network revealer (Tang, 2026w), the triple-demotion completer (Tang, 2026x), and the theoretical extension and methodological output (this paper)—marks the complete deployment of the Factor Hierarchy Law in a sixth discipline.

Data Availability Statement

This paper is a theoretical-extension and system-integration paper. All empirical foundations have been reported in detail in Tang (2026v), Tang (2026w), and Tang (2026x). The supplementary analyses of this paper (quantification of regime-factor semi-continuity, identification of the binary structure of the confirmed-breakpoint network, and the WLS versus enrichment-zone contrast) are based on re-analyses of empirical data already published in the preceding papers. The data are completely public, and any researcher can independently reproduce all results.

References

- [1] Bai, J., & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78.
- [2] Bethe, H. A., & Bacher, R. F. (1936). Nuclear Physics A. Stationary States of Nuclei. *Reviews of Modern Physics*, 8(2), 82–229.

- [3] Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605.
- [4] Mayer, M. G., & Jensen, J. H. D. (1955). *Elementary Theory of Nuclear Shell Structure*. Wiley.
- [5] Onsager, L. (1944). Crystal statistics. I. A two-dimensional model with an order-disorder transition. *Physical Review*, 65(3–4), 117–149.
- [6] Strutinsky, V. M. (1967). Shell effects in nuclear masses and deformation energies. *Nuclear Physics A*, 95(2), 420–442.
- [7] Tang, S. (2026a). Dual Anchors: China-U.S. Factors Jointly Determining Global Gold and Crude Oil Pricing — Systematic Evidence Based on Exhaustive Testing with VIF Constraints. Zenodo, DOI: 10.5281/zenodo.21091486.
- [8] Tang, S. (2026b). Interest Rates Without an Anchor: Multi-Window Percentiles, Exhaustive Testing, and the Pricing Logic of the China-U.S. Yield Spread—On the Dual-Anchor Regime and Institutional Interface Theory of Global Asset Pricing. Zenodo, DOI: 10.5281/zenodo.21096496.
- [9] Tang, S. (2026c). The Unified Field of Spreads: The China-U.S. Yield Spread as a Regime Variable in Global Asset Pricing—On the Hierarchical Structure of Institutional Interfaces and Supplements to Interest Rate Parity, the Triffin Dilemma, and Related Theories. Zenodo, DOI: 10.5281/zenodo.21096549.
- [10] Tang, S. (2026d). The Anchor Shift: Reconstructing Global Exchange Rate Determination — The China-U.S. Yield Spread as a Regime Variable, with Systematic Alternatives to Interest Rate Parity, the Mundell-Fleming Model, and the Triffin Dilemma. Zenodo, DOI: 10.5281/zenodo.21098055.
- [11] Tang, S. (2026e). The Hierarchy of Factors: How Regime Variables Reset Market Pricing—Evidence from Exhaustive Testing Across 15 Global Equity Markets for a Factor Hierarchy Theory. Zenodo, DOI: 10.5281/zenodo.21099589.
- [12] Tang, S. (2026f). The Regime Switch of Capital Flows: How the China-U.S. Yield Spread Drives Global Capital Reallocation in a Tiered Manner—Exhaustive Evidence Based on TIC Panel Data, ETF Flows, and China's Cross-Border Data. Zenodo, DOI: 10.5281/zenodo.21100341.
- [13] Tang, S. (2026g). Testability: Factor Hierarchy Theory and Scientific Methodology for Multi-Factor Complex Systems — Complete Empirical Evidence from Global Asset Pricing and a Cross-Disciplinary Application Paradigm. Zenodo, DOI: 10.5281/zenodo.21104307.
- [14] Tang, S. (2026h). Planetary Radius as a Regime-Dependent Relational Quantity: Evidence for a Paradigm Defect in Exoplanet Radius Measurement—A Cross-Disciplinary Empirical Test in Astronomy Based on Factor Hierarchy Theory and the Four-Test Method. Zenodo, DOI: 10.5281/zenodo.21105475.
- [15] Tang, S. (2026i). Self-Consistency Rupture in Stellar Atmosphere Models: The Second Astronomical Empirical Case for the Factor Hierarchy Law — Evidence from Internal Cross-Validation Based on the Four-Test Method. Zenodo, DOI: 10.5281/zenodo.21127665.

[16] Tang, S. (2026j). The Stellar Main-Sequence Regime Boundary: Five-Dimensional Independent Verification of the 4762 K Critical Point and the Systematic Establishment of the Factor Hierarchy Law — Cross-Dimensional Integrated Evidence Based on the Four-Test Method. Zenodo, DOI: 10.5281/zenodo.21128329.

[17] Tang, S. (2026k). Information-Theoretic Duality Between Regression Topology and Statistical Mechanics—A Data-Driven Equivalence of the Ehrenfest Classification Validated on the 2D Ising Model as the Physics Cornerstone for the Factor Hierarchy Law. Zenodo, DOI: 10.5281/zenodo.21154577.

[18] Tang, S. (2026l). Double-Peak Fission: The Hidden Structure of Finite-Lattice Phase Transitions in the 3D Ising Model — Verification of the Factor Hierarchy Law on an NP-Complete System: Double-Peak Fission, Transition Regime, Piecewise Control Function, and Multi-Regime State Decoupling Algorithm. Zenodo, DOI: 10.5281/zenodo.21257777.

[19] Tang, S. (2026m). The Reduction of Universality: A Regime Boundary Theory for Finite-System Phase Transitions — Cross-Model Evidence for Bimodal Splitting, the Precursor Regime Boundary, and the Tool-Dependence Theorem. Zenodo, DOI: 10.5281/zenodo.21258474.

[20] Tang, S. (2026n). Regime Audit of the Standard Model of Particle Physics by the Factor Hierarchy Law — Activation Operators, Dual-Center Regimes, and Precision Locking of Charm/Bottom Quark Thresholds(I). Zenodo, DOI: 10.5281/zenodo.21276887.

[21] Tang, S. (2026o). Strange Region Discovery, Precursor Cascade, and the Transition Zone Width Theorem — Four-Tool Cross-Validation, Cascade Asymmetry Confirmation, and Mathematical Proof of the Switching Type Dichotomy(II). Zenodo, DOI: 10.5281/zenodo.21277300.

[22] Tang, S. (2026p). Gauge Group Structure Theorem, Direct Product Theorem, and the Three-Generation Proposition — The Inevitable Emergence of $SU(3) \times SU(2) \times U(1)$ from Regime Boundaries and the Topological Constraint on the Number of Heavy Quarks(III). Zenodo, DOI: 10.5281/zenodo.21291670.

[23] Tang, S. (2026q). The Demotion of Gauge Field Theory, the Theoretical Reconstruction of the Standard Model, and the Five-Layer Axiomatization Structure — Precursor Cascade Asymmetry, the Regime Function of the Higgs Mechanism, the Statistical Correspondence of Virtual States, and the Final Formalization of the Factor Hierarchy Law (IV). Zenodo, DOI: 10.5281/zenodo.21292223.

[24] Tang, S. (2026r). From the Law of Meta-Laws to an Operating System for Scientific Discovery: The Axiomatized Core, Philosophical Foundation, and Testability Norms of the Factor Hierarchy Law — The Establishment of Humanity's First Proceduralized System for Scientific Discovery. Zenodo, DOI: 10.5281/zenodo.21306949.

[25] Tang, S. (2026s). The Calcium-Lanthanum Singularity Theorem: Exhaustive Discovery and Multi-Tool Cross-Validation of the d-Block and f-Block Regime Switching Centers in the Periodic Table — First

Deployment of the Factor Hierarchy Law in Chemistry, Verification of the Statistical Signature of Regime Factors, and Confirmation of the Chemical $\times$ -Blindness (I).Zenodo,DOI:10.5281/zenodo.21403322

[26] Tang, S. (2026t). The Regime Periodic Table: Regime Identity Cards for 118 Elements and the Reduction of Universality of the Periodic Law — Elimination of Exceptions, Numerical Predictions for Elements 119/120, and Cross-Sample Convergence Audit (II).Zenodo,DOI:10.5281/zenodo.21403594

[27] Tang, S. (2026u). From Regime Boundaries to Chemical Gauge Group: The Inevitable Emergence of the Periodic Law and the Cross-Disciplinary Unification of the Factor Hierarchy Law — Derivation of the Chemical Gauge Group Theorem, the Chemical Counterpart of the Higgs Mechanism, and the Establishment of the Five-Layer Axiomatic Structure (III).Zenodo,DOI:10.5281/zenodo.21403811

[28] Tang, S. (2026v). Nuclear Shell Regime Audit: The Magic-Number Shift Theorem and the Discovery of the N=14 Statistical Rupture——The η Hierarchy Structure, Bai-Perron Optimal 10 Breakpoints from Zero-Preset Exhaustive Search, and Six Quantitative Theorems.Zenodo, DOI: 10.5281/zenodo.21621800.

[29] Tang, S. (2026w). Regime-Switch Network and the True Structure of the Nuclear Shell: Internal Self-Consistency of the Confirmed-Breakpoint Network, Completeness Audit of the Shell Model, and Fifteen New Discoveries——Transition-Zone Width Theorem, Cascade Skeleton, Liquid-Drop Masking Effect, and Shell-Correction Compensation Traces Based on Multi-Tool Aggregation.Zenodo, DOI:10.5281/zenodo.21622347

[30] Tang, S. (2026x). The Triple Demotion of the Nuclear Shell: The Shell Model, the Liquid Drop Model, and the Independent-Particle-Motion Axiom —— Derivational Demotion Based on 2D Ising Exact Solution Calibration, Continuous Sphericity Weight Recovery, and the QCTI Index.Zenodo, DOI:10.5281/zenodo.21634385

[31] Yang, C. N. (1952). The spontaneous magnetization of a two-dimensional Ising model. *Physical Review*, 85(5), 808–816.